

Climate Livestock Dynamics in Somalia

Rainfall Trends and Shifts in Pastoral Livestock Stock Across Somalia (1981–2024)

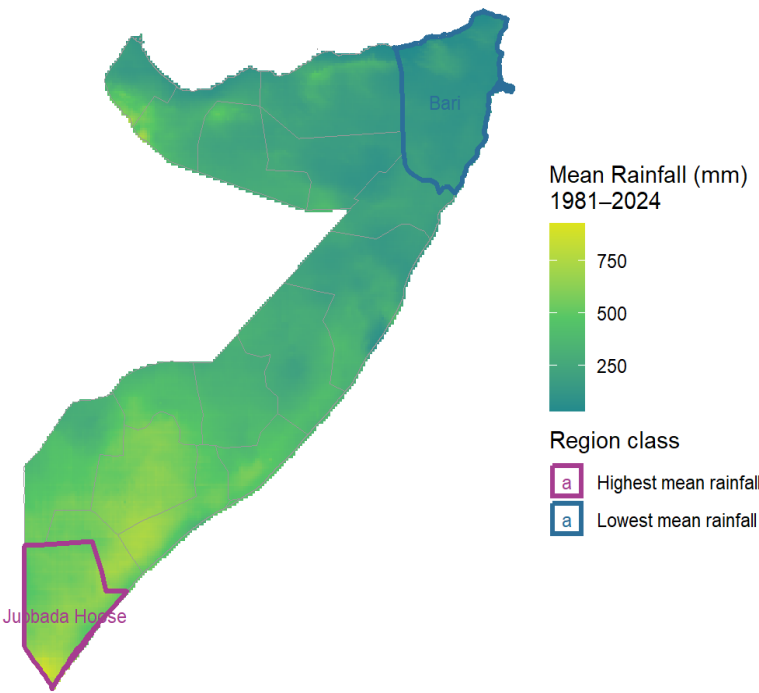
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The Environmental Issue: Climate Variability Threatens Somalia’s Livestock Economy

- Livestock account for **80% of exports** and underpin pastoral livelihoods.
- Somalia experiences extreme rainfall variability and frequent droughts.
- Environmental shocks destabilize pasture availability and herd survival.

Somalia Mean Rainfall (1981–2024): Wettest vs Driest Region

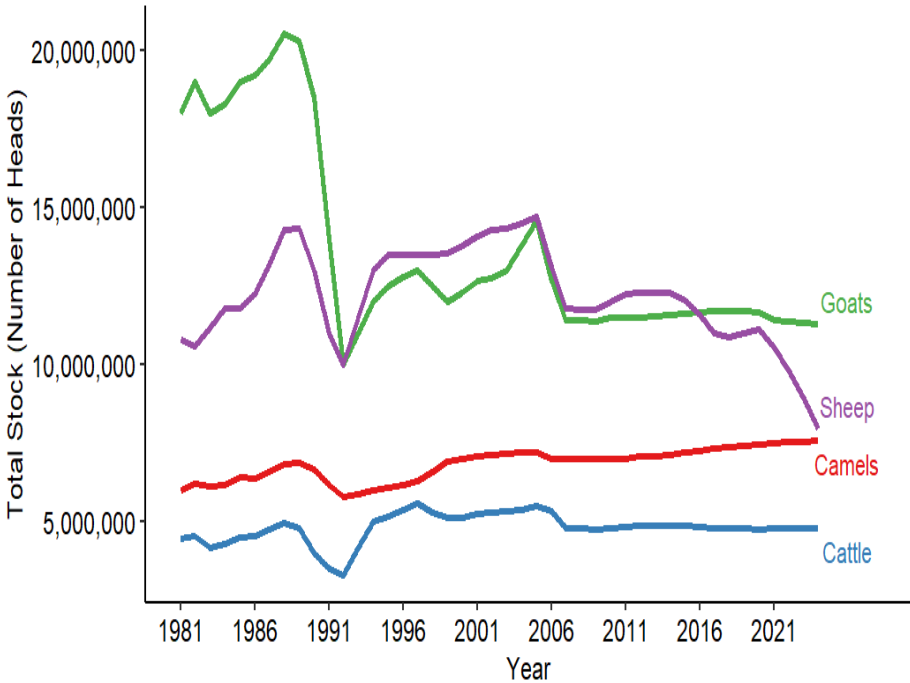


Data & Methods

- CHIRPS satellite rainfall (1981–2024) combined with FAOSTAT livestock data.
- R spatial analysis used to map rainfall variability across Somalia.
- Generalized Additive Models (GAMs) used to test climate–livestock relationships.

Livestock Population Trends in Somalia

Annual headcount (1981–2024)

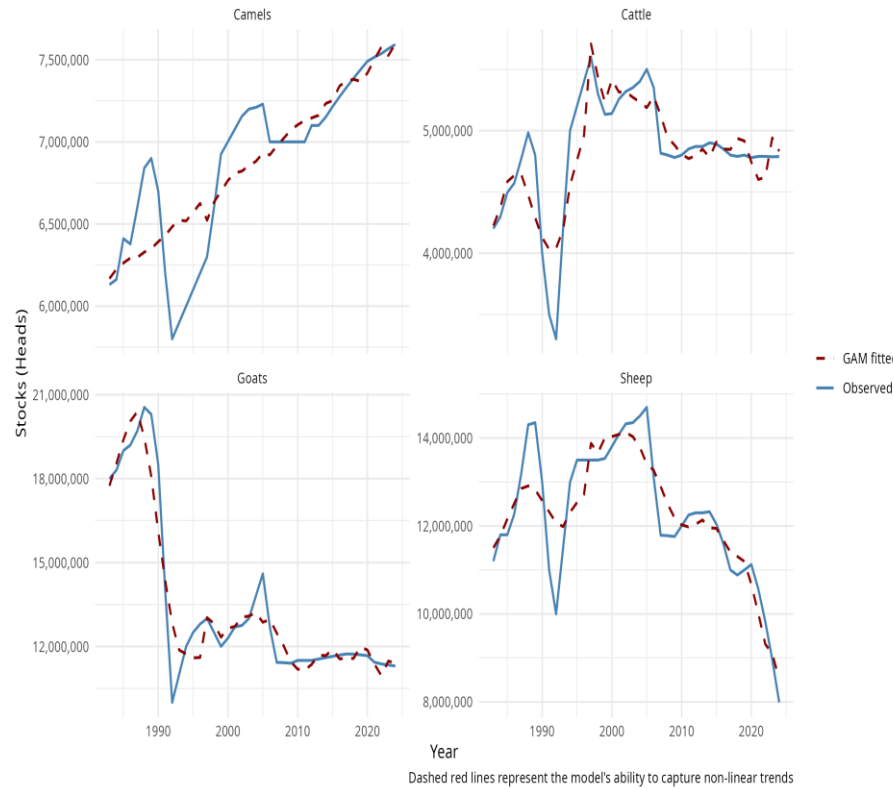


Source: FAOSTAT | Yale Environmental Data Science

Key Finding/ Insights

- Rainfall shows no long-term trend, but strong year-to-year variability.
- Livestock respond differently to climate variability:
 - Camels increasing
 - Goats declining
 - Cattle relatively stable

Model Performance: Observed vs. Predicted Stocks
GAM results incorporating rainfall anomaly lags (0-2 years)



Interpretation / Implications

- Livestock trends reflect climate plus markets, conflict, and pastoral mobility.
- National rainfall averages may hide important regional climate impacts.
- Future work should integrate regional climate indicators and socio-economic data.

Annual Rainfall Anomalies in Somalia (1981–2024)

Highlighting 5-year intervals for improved readability

